UNITED INTERNATIONAL UNIVERSITY (UIU)
Dept. of Computer Science & Engineering
Trimester: Summer 2026 (Mid)
Course No: CSE 3812 **Title:** Artificial Intelligence Laboratory
Section: H **Time:** 60 mins **Marks:** 30

Instructions:

- While answering the questions make sure to follow this convention,
  - First create a new folder in any directory on your current PC where you want to write your code. The folder name will be your rollNumber_Mid e.g. for me, it will be 1907016_Mid
  - Then go into the folder. The folder will contain both the python scripts as answers to the two questions provided.
  - Suppose for me, my roll was 1907016. So, the first filename should be `1907016_a.ipynb` and the second one should be `1907016_b.ipynb`. For you, only the roll number part will change. `[If someone wants to write in a (dot)py script, that's completely fine too]`
  - Properly save the scripts at regular intervals if you're coding on an offline basis because unlike google colab, offline codes sometimes don't have autosave option on.
  - If you want to code online in Google colab, then you should create a new notebook after going into https://colab.research.google.com/ following the same name convention.
  - After you're done with the coding part, you should download the notebook from the `File` option on the top left corner as an ipynb file in the folder directory that I instructed as before.
- If you followed the instructions properly, now your folder directory has two ipynb/py files in it.
- Then, upload the whole folder in this Link.

GitHub Repo Link: https://github.com/IceColdMartini/262-AI-Lab-H

Go to the Repo, read the `readme.md` and you will find every answer to all your questions.

Aliceland Relief Rover

Allocated Marks: 15

Following severe seasonal flooding in the territory of ***Aliceland***, emergency response forces are deploying an autonomous relief rover to navigate from a central distribution depot to a stranded aid station. The terrain across Aliceland is non-uniform: paved roads allow rapid transit, muddy tracks impede progress, flooded marshes incur high movement resistance, and floodwalls block passage entirely.

The territory is digitized as a 2D integer matrix **`grid`** of dimension **`RxC`**. The numerical value stored within each cell **`grid[r][c]`** dictates the movement mechanics:

- `-1`: Impassable Floodwall (The rover cannot enter this coordinate).
- `1`: Paved Road (Movement cost $g_{step} = 1.0$).
- `3`: Muddy Track (Movement cost $g_{step} = 3.0$).
- `5`: Flooded Marsh (Movement cost $g_{step} = 5.0$).

The rover can only move in **4 cardinal directions (Up, Down, Left, Right)**. When transitioning from a current cell to an adjacent cell `(nr, nc)`, the path cost increases by the traversal cost of the destination cell, such that

$$g(n_{\text{neighbor}}) = g(n_{\text{current}}) + \text{grid}[nr][nc]$$

Your goal is to modify the provided baseline function `base_astar_search` to implement `solve_aliceland_rover(grid, start, goal)`. The function must identify the optimal path minimizing cumulative energy cost while avoiding all floodwalls.

A helper code has been provided in the github directory where almost everything is filled up except for the part which you are supposed to modify. If you test your code by executing Cell-5 of the ipynb on the sample data, it will give you an output like this:

```
Discovered Path: [(0, 0), (0, 1), (1, 1), (1, 2), (1, 3), (2, 3), (3, 3)]
Total Traversal Cost: 8.0
```

Karp City Hovercraft

Allocated Marks: 15

In the metropolis of **Karp City**, medical supply hovercrafts transport vaccines across a city grid containing `roadways(0)` and `high-rise structures(1)`. While standard movement occurs across adjacent cardinal road cells at unit cost (1.0), Karp City features **Quantum Warp Portals** to bypass urban blockades.

A warp portal connects an entry cell (r_1, c_1) directly to an exit cell (r_2, c_2) with a fixed warp energy cost $E_{warp} = 1.0$. Portals are configured via a Python dictionary mapping entry coordinates to exit coordinates:

```
portals = {
    (0, 1): (0, 3),   # Teleports from (0, 1) directly to (0, 3)
    (2, 0): (4, 4)    # Teleports from (2, 0) directly to (4, 4)
}
```

When evaluating successor states from coordinate **`(r, c)`**, the search engine must generate two sets of candidate moves:

1. Standard cardinal movements to adjacent unblocked road cells where **`grid[nr][nc] == 0`**, with step cost `1.0`.
2. If `(r, c)` matches a key in portals, a teleportation move to `portals[(r, c)]` with step cost `portal_cost` `(1.0)`.

Modify the baseline A* algorithm to implement **`solve_karp_city(grid, start, goal, portals, portal_cost=1.0)`**

A helper code has been provided in the github directory where almost everything is filled up except for the part which you are supposed to modify. If you test your code by executing Cell-5 of the ipynb on the sample data, it will give you an output like this:

```
Hovercraft Path: [(0, 0), (0, 1), (0, 3), (1, 3), (2, 3), (2, 4), (3, 4), (4, 4)]
Total Traversal Cost: 7.0
```

HAPPY CODING

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